Project Wonky Sprout: A Gamified Assistive Technology Development Roadmap for Neurodivergent Self-Regulation and Communication Bridging

1. Executive Strategic Alignment and Operator Profiling

1.1 Introduction

The "Wonky Sprout" initiative represents a paradigmatic shift in the development of Assistive Technology (AT). Rather than approaching the fabrication of sensory tools as a medical necessity or a purely functional engineering challenge, this roadmap utilizes a **Gamified Engineering Framework**. This approach is specifically tailored to the psychographic and professional profile of the primary user—herein referred to as the **Operator**—to bridge the psychological gap between intent and execution.

The project addresses two critical functional deficits identified in the preliminary user analysis: the dissonance between internal emotional states and external facial affect (a common trait in Autism Spectrum Disorder), and the susceptibility to sensory overload in uncontrolled environments. By synthesizing the Operator's profound professional expertise in industrial electrical systems with the motivational structures of Role-Playing Games (RPGs), "Wonky Sprout" transforms the rehabilitation process into a structured, technical campaign of self-augmentation.

1.2 The Operator Persona: A Competency Analysis

To design an effective development roadmap, one must first understand the "Player Character." The Operator is not a novice; the provided documentation reveals a highly skilled technician with a GS-12 level proficiency in complex electrical systems.[1] This background is the cornerstone of the gamification strategy. The Operator does not require simplified instructions; rather, they require a translation of therapeutic goals into their native dialect: **Industrial Engineering Standards**.

The analysis of the Operator's resume reveals a specific set of high-level competencies that will be repurposed as "In-Game Abilities":

Professional Competency	In-Game Translation (RPG Attribute)	Application in Wonky Sprout
Root Cause Analysis	**Debug Mode (Intelligence)**	Diagnosing social miscommunications or sensory triggers as "system faults" rather than personal failures.
VFR (Voltage Frequency Regulator) Maintenance	**Regulation (Stamina)**	Managing emotional "voltage" and "frequency" to prevent burnout (system overload).
Objective Quality Evidence (OQE) Generation	**XP Generation (Progression)**	Using documentation and self-reporting as the metric for leveling up.
Instruction Card / SOP Creation	**Scripting (Wisdom)**	Creating "macro" responses for social situations, effectively automating charisma.
Cable Encapsulation / VPI	**Crafting (Dexterity)**	Fabricating durable, weather-sealed wearable

Systems		electronics using industrial techniques.

The Operator's correspondence with **Tools for Life** [1] highlights a critical psychological driver: the desire to transition from a passive consumer of technology to an active creator. The explicit mention of "intense, preoccupying interests in 3D printing, CAD, and electronics" suggests that the fabrication process itself is a mechanism for regulation. Therefore, the roadmap focuses heavily on the *act* of building as much as the final product.

1.3 The "Nonverbal Mismatch" Anomaly

The primary communicative deficit identified is the misalignment between the Operator's facial expressions and their inner feelings.[1] In typical social interactions, this "Nonverbal Mismatch" leads to listener confusion and Operator frustration. From a systems engineering perspective, this is akin to a **Sensor/Display Desynchronization**. The internal sensor (the brain) registers "Joy," but the external display (the face) reads "Neutral" or "Angry."

The proposed solution—the **Sensory Helmet**—functions as an auxiliary display unit. It bypasses the faulty biological wiring and allows the Operator to manually or semi-automatically drive the external signal, ensuring data integrity in social transmissions. This is not merely a mask; it is a calibration tool, aligning the "Output" with the "Input" in a manner consistent with the Operator's experience calibrating 500kW motor generators.[1]

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2. The Gamification Engine: "Project Sprout" Mechanics

This section outlines the "Game Rules" that govern the development process. By imposing a game layer over the engineering tasks, we utilize the dopamine feedback loops inherent in RPGs to maintain motivation through the tedious phases of fabrication and testing.

2.1 The Experience Point (XP) System: Validating "Objective Quality Evidence"

In the Operator's professional life at the Trident Refit Facility, work is not considered complete until "Objective Quality Evidence" (OQE) is gathered to certify maintenance.[1] This roadmap adopts the OQE standard as the currency for progression. "XP" is not awarded for *trying*; it is awarded for *verifiable completion* documented in a "Quest Log" (Project Journal).

The XP thresholds are calibrated to incentivize both the "Grind" (repetitive tasks) and the "Boss Battles" (complex integration challenges).

Activity Class	Engineering Task Example	XP Value	Verification Artifact (OQE)
The Grind (Farming)	Post-processing 3D prints (sanding/priming); Soldering header pins to microcontrollers.	50 XP / Hour	Photographic evidence of finished parts; Continuity test results.
Skill Check	Designing a PCB enclosure in CAD; Writing a Python script for LED control.	300 XP	STL file generation; GitHub commit hash.
Side Quest	Networking with **TOM Global** or **Tools for Life**.[1]	500 XP	Email thread initiation; Meeting notes.
Boss Battle	"Black Band" testing the helmet under full load; Successful public field test.	2000 XP	Video documentation; System log analysis showing no thermal throttling.
Guild Contribution	Volunteering 3D printing skills for external requests.[1]	1000 XP	Delivery of printed parts to the requesting agency.

2.2 The Skill Trees: A Roadmap for Self-Augmentation

The Operator has expressed a specific desire to learn Artificial Intelligence (AI) and coding while leveraging existing strengths in fabrication.[1] The Skill Tree mechanic visualizes this trajectory, breaking down the overwhelming goal of "Building an AI Helmet" into discrete, unlockable nodes.

2.2.1 Tree A: The Artificer (Hardware & Fabrication)

This tree leverages the Operator's "Mastery" level skills in industrial repair. The progression moves from basic assembly to advanced material science.

- **Tier 1 (Apprentice):** Calibration of FDM printers for dimensionally accurate structural parts.
- **Tier 2 (Journeyman):** Integration of "Cable Encapsulation" techniques [1]—using silicone potting compounds to waterproof the helmet's electronics, mimicking the sealed motor windings the Operator is famous for.
- **Tier 3 (Master):** Advanced ergonomics—using 3D scanning to create a custom-fitted "TPU Liner" that rivals the comfort of industrial PPE.

2.2.2 Tree B: The Technomancer (Software & AI)

This tree addresses the learning curve for coding. It starts with the basics of logic control (familiar to someone who knows VFR circuits) and escalates to machine learning.

- **Tier 1 (Apprentice):** "Hello World" on an LED matrix; Basic state machine logic (If Button A -> Show Smile).
- **Tier 2 (Journeyman):** Integrating I2C sensors (accelerometers, microphones) to create reactive lighting effects.
- **Tier 3 (Master):** Deploying a TensorFlow Lite model on an ESP32 microcontroller to analyze voice tone and suggest emotion icons.

2.2.3 Tree C: The Diplomat (Social Protocols)

This tree gamifies the soft skills of communication, turning social interactions into "encounters" with win states.

- **Tier 1 (Apprentice):** Using the helmet to signal "Overload" instead of verbally shutting down.
- **Tier 2 (Journeyman):** Using the helmet to initiate a conversation (e.g., displaying a "Hello" animation).
- **Tier 3 (Master):** "Parallel Play"—collaborating with the **Georgia Tech TOM Global** chapter [1] on a shared build, effectively navigating a complex social environment using the role of "Technical Expert" as a shield.

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3. System Architecture: The Dual-Core Ecosystem

The technical scope of "Wonky Sprout" is bifurcated into two complementary hardware environments: the **Mobile Unit (Helmet)** and the **Static Unit (Sensory Home)**. This mirrors the Operator's professional environment, where submarines (Mobile Units) require a Refit Facility (Static Unit) for maintenance and support.[1]

3.1 The Mobile Unit: The "Wonky Sprout" Sensory Helmet

The helmet is the Operator's "Exosuit." It provides a buffered interface with the chaotic outside world.

3.1.1 Structural Integrity and Materials

Drawing on the Operator's experience with "sealed motor windings" [1], the helmet must be robust. We reject standard PLA filament in favor of **ASA (Acrylonitrile Styrene Acrylate)**. ASA offers the UV resistance of industrial plastics with the printability of ABS, crucial for a device worn outdoors. The internal structure will utilize **TPU (Thermoplastic Polyurethane)** printed with gyroid infill, functioning as a shock absorber similar to the "propulsion bearings" the Operator maintains.[1]

3.1.2 The "VFR" Power System

The Operator works with Voltage Frequency Regulators.[1] The helmet requires a miniaturized version of this logic.

- **Power Source:** High-discharge 18650 Li-Ion cells (3S or 4S configuration).
- **Regulation:** A dedicated Buck Converter system stepping down voltage to stable 5V and 3.3V rails. This ensures that even if the battery voltage sags under load (e.g., running all LEDs at 100% brightness), the microcontroller (the "Brain") receives clean power.
- **Safety:** A BMS (Battery Management System) is non-negotiable. It monitors cell balancing and prevents thermal runaway—a direct application of the Operator's "Safety First" workplace culture.[1]

3.1.3 The Communication Array (The "Face")

To solve the nonverbal mismatch, the helmet features a flexible **WS2812B LED Matrix** (8x32 or 16x16) mounted behind a smoked acrylic visor.

- **Function:** It renders 8-bit emote icons.
- **Input:** A wrist-mounted "Macropad" (a small mechanical keyboard) connects wirelessly via ESP-NOW protocols.
- **Logic:** The Operator presses a tactile switch corresponding to their *internal* state. The helmet displays the corresponding *external* signal. This restores agency: the Operator controls how they are perceived.

3.2 The Static Unit: The "Dry Dock" Sensory Home

The home environment functions as the "Regeneration Station." Just as a submarine returns to Kings Bay for refit, the Operator returns home to lower sensory "heat."

3.2.1 Automated Decompression Protocols

Utilizing **Home Assistant** running on a local server (Raspberry Pi), the home detects the helmet's Bluetooth beacon upon arrival.

- **The "Welcome Home" Script:**
  - **Visuals:** All high-Kelvin overhead lights (4000K+) fade to zero. Low-Kelvin ambient strips (1800K, Amber) fade up to 30%.
  - **Audio:** A "Soundscaping" system (Sonos or open-source equivalent) crossfades from silence to "Brown Noise," masking erratic street sounds.
  - **Climate:** HVAC fans adjust to a constant "White Noise" RPM, avoiding the sensory spike of cycling on and off.

3.2.2 The Workshop as "Bridge"

The Operator's resume indicates proficiency with "CITRIX, LDS, and ATIS" [1]—complex database and tracking systems. The home workshop will feature a digital "Mission Control" dashboard.

- **Displays:** Monitors showing the status of 3D prints (OctoPrint), local weather (barometric pressure triggers migraines for some), and a "Task Board" (Kanban style) tracking the "Wonky Sprout" build progress.

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4. Phase I: The "Keel Laying" (Helmet Development Roadmap)

This phase focuses on establishing the physical and electronic architecture of the Mobile Unit.

4.1 Quest 1: The Chassis (CAD & Fabrication)

Objective: Fabricate the helmet shell.
Prerequisites: 3D Printer, ASA Filament, Calipers.
Resume Link: The Operator is skilled in "taking precision measurements on shaft bearing journals".1 This skill is critical for measuring head circumference and creating a dimensionally accurate CAD model.
The Operator will begin by photogrammetrically scanning their head (using open-source tools

like Meshroom). This scan is imported into CAD software (Fusion 360 or SolidWorks). The helmet shell is modeled around this "Keep-Out Zone," ensuring a perfect fit—critical for long-term wearability and preventing sensory friction.

Victory Condition (OQE): The printed shell fits comfortably for 60 minutes without pressure points.

4.2 Quest 2: The Nervous System (Electronics Integration)

Objective: Create the wiring harness.
Prerequisites: Soldering Station, Wire Strippers, Heat Shrink.
Resume Link: The Operator is an expert in "cable encapsulation" and "VPI systems".1
The wiring harness must be modular. Connectors (JST-XH or XT60) should be used rather than direct soldering to components, allowing for easy "Field Repair" or upgrades. The Operator will apply their knowledge of "insulation resistance" [1] to ensure no short circuits occur between the power rails and the sensitive data lines of the microcontroller.

Victory Condition (OQE): A "Smoke Test" is passed. The system powers up, LEDs cycle through a test pattern (Red/Green/Blue), and current draw remains within calculated limits.

4.3 Quest 3: The "Black Band" Stress Test

Objective: Validate system stability under load.
Resume Link: The Operator performs "Black Band" testing on motor generators.1 This concept is adapted for the helmet.
We do not simply assume the code works. We stress it.

- **The Test:** Run the LED matrix at 100% White brightness (maximum current draw) while simultaneously flooding the microcontroller with Wi-Fi packets and button interrupts.
- **The Metric:** Monitor the voltage sag on the 5V rail. If it drops below 4.5V, the wiring is too thin or the battery C-rating is too low.
- **The Fix:** If the test fails, perform "Root Cause Analysis" [1]—is it a thermal throttle? A bad solder joint?

Victory Condition (OQE): The system runs the "Stress Macro" for 1 hour without crashing or overheating (>50°C).

5. Phase II: The "Synthetic Mind" (Software & AI Integration)

This phase addresses the Operator's desire to learn coding and AI.[1]

5.1 Quest 4: The Logic Gate (Firmware Basics)

Objective: Program the ESP32 Microcontroller.
Context: The Operator understands logic gates from "SCR firing circuits" in VFRs.1
The firmware will be written in C++ (Arduino Framework) or MicroPython. It follows a strict "State Machine" architecture:

- **State IDLE:** Display "Neutral" animation. Low power consumption.
- **State ACTIVE:** Display selected "Emotion."
- **State OVERLOAD:** Triggered by a panic button. Displays "Do Not Disturb" and activates noise-canceling routines.

5.2 Quest 5: The AI Co-Pilot (Advanced)

Objective: Implement assistive AI.
Context: "Artificial Intelligence (AI) and how it can be used to develop personalized, structured tools".1
We will integrate a **TinyML** (Machine Learning for Microcontrollers) component.

- **Voice Analysis:** Using a PDM microphone, the ESP32 samples the Operator's speech. A pre-trained TensorFlow Lite model analyzes the *prosody* (tone/rhythm) rather than the words.
- **Application:** If the model detects "distressed" vocal patterns (elevated pitch, rapid pace), it automatically cues the helmet to display a "Please be patient" icon, even if the Operator forgets to press the button. This creates a safety net.

6. Phase III: The "Guild" Integration (Community & Funding)

The Operator's email to Tools for Life [1] reveals a crucial resource: the willingness to volunteer. This is not just altruism; it is a strategic asset for acquiring knowledge and resources.

6.1 Quest 6: The "Tools for Life" Connection

Objective: Leverage the State AT Program.
Contact: Hunter McFeron (Acquisition Services Manager).
Strategy:

- **Device Loans:** While Tools for Life does not do assessments [1], they do loan devices. The Operator should borrow various switches, buttons, and mounting arms to "beta test" ergonomics before buying expensive components for the helmet.
- **Volunteering:** The Operator offered 3D printing skills. By fulfilling print requests for the program, the Operator gains "Reputation Points" and potentially access to scraps/surplus materials or introductions to other engineers.

6.2 Quest 7: The "TOM Global" Alliance

Objective: Join the Maker Community.
Contact: Georgiatech@tomglobal.org.1
Strategy:

- **The Makeathon:** TOM (Tikkum Olam Makers) organizes "Makeathons" where engineers build AT for people with disabilities.
- **The Role:** The Operator should join a team not as a "patient," but as a "Lead Fabricator." Their resume—specifically the ability to "oversee journeyman electricians" and "coordinate with engineering departments" [1]—makes them an invaluable team lead.
- **Benefit:** This provides access to Georgia Tech's advanced fabrication labs (laser cutters, resin printers) which may be needed for the finer details of the Wonky Sprout project.

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7. Quality Assurance: The "Instruction Card" Protocol

The Operator's resume highlights the importance of "Instruction Cards" and "QA Forms".[1] In the context of Autism, these are **Cognitive Prosthetics**.

7.1 The "Meltdown" SOP (Standard Operating Procedure)

When the Operator is in a high-stress state, executive function (planning/memory) degrades. We must engineer a failsafe.

- **The Artifact:** A physical, laminated card (or an e-ink display on the wrist) detailing the "Emergency Shutdown Procedure."
- **Content:**
  1. "Stop Moving."
  2. "Press Blue Button on Helmet (Activates ANC)."
  3. "Check Heart Rate."
  4. "Navigate to [Home Coordinates]."
- **Why it works:** It removes the need to *think*. The Operator simply follows the trusted SOP, just as they would follow a maintenance card during a submarine drill.

7.2 The "OQE" Project Journal

To maintain the gamification, every milestone must be recorded.

- **Format:** A Markdown-based wiki or a physical logbook.
- **Entries:** Must include photos of the build, snippets of code, and "Post-Mortem" analysis of any failures.
- **Psychological Benefit:** This serves as concrete proof of competence. When the Operator feels overwhelmed or "broken," the OQE Journal provides irrefutable evidence of their ability to build, solve, and adapt.

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8. Conclusion: From Technician to Architect

The "Wonky Sprout" roadmap is more than a build guide; it is a restructuring of the Operator's relationship with their own neurodivergence. By reframing the challenges of ASD—sensory overload, communication mismatch, executive dysfunction—as engineering constraints within a complex system, we empower the Operator to solve them using their strongest assets.

The Operator is not a passive recipient of care. They are a **GS-12 Engineering Technician** capable of maintaining the propulsion systems of nuclear submarines. They are more than qualified to build their own life support system.

The campaign begins now.

Status: Green Board.
Authorization: Granted.
Next Action: Initiate Quest 1 (The Chassis).

Appendix A: Technical Specifications and Bill of Materials (BOM)

This appendix provides the granular data required to execute the fabrication phases, formatted as "Loot Tables" for the gamified context.

A.1 The Mobile Unit (Helmet) - Component List

Component Class	Item Description	Gamified Rarity	Sourcing Strategy
Main Processor	**ESP32-S3 DevKitC** (Dual Core, Wi-Fi/BT LE).	Rare	Purchase online. Look for "S3" variant for AI instruction sets.
Display Matrix	**WS2812B 8x32 Flexible Panel**.	Uncommon	Available from hobbyist electronics

			suppliers.
Power Cell	**Samsung 35E 18650 Li-Ion** (3500mAh).	Epic	Salvage from high-end laptop batteries (requires *Safety Check*) or purchase new.
Chassis Material	**Prusament ASA** (Galaxy Black).	Legendary	Purchase. Requires enclosed printer to avoid warping.
Liner Material	**TPU 95A** (Shore Hardness).	Rare	Purchase. Use 5% Gyroid infill for "Air Ride" suspension.
Input Device	**Mechanical Switch (Cherry MX Blue)**.	Common	Salvage from old keyboards. "Clicky" feedback confirms input without looking.
Waterproofing	**Silicone Conformal Coating**.	Epic	Use "Cable Encapsulation" skills [1] to brush onto PCB.

A.2 The Static Unit (Home) - Automation Stack

Module	Software/Hardware	Function
Core Server	**Home Assistant OS** (running on Raspberry Pi 4).	The "Central Nervous System" of the house.
Protocol	**MQTT (Message Queuing**	Lightweight messaging

	Telemetry Transport).	protocol for the Helmet to talk to the House.
Lighting	**Zigbee Smart Bulbs** (Philips Hue or equivalent).	Low-latency control. Requires a Zigbee USB dongle for the server.
Presence	**ESP32 Bluetooth Beacon**.	The Helmet broadcasts a specific BLE UUID. The House scans for it to trigger "Welcome" scripts.

A.3 Code Snippet: The "Emotion Engine" (Pseudo-Code)

C++

```
// Wonky Sprout Firmware v1.0
// Module: Emotion State Machine

#include <FastLED.h>

// Define State Constants
enum State {
  STATE_NEUTRAL,
  STATE_JOY,
  STATE_FOCUS,
  STATE_OVERLOAD
};

State currentState = STATE_NEUTRAL;

void loop() {
  // Read Input (Debounced Buttons)
```

```
  if (btn_Joy.pressed()) {
    currentState = STATE_JOY;
    updateDisplay(ICON_SMILE, CRGB::Green);
  }

  if (btn_Overload.pressed()) {
    currentState = STATE_OVERLOAD;
    updateDisplay(ICON_STOP, CRGB::Red);
    activateHaptics(); // Vibrating alert
    sendMQTT("home/scene/recovery"); // Tell House to go dark
  }

  // Safety Watchdog
  checkBatteryLevels();
}
```

A.4 The "Quest Log" Template (Project Management)

The Operator should utilize a Kanban board (Trello or GitHub Projects) to track "Quests."

- **Backlog (The Quest Board):** All future ideas (e.g., "Add Night Vision," "Train AI Model").
- **In Progress (Active Quests):** Tasks currently on the workbench. Limit to 3 to prevent executive dysfunction.
- **Blocked (Awaiting Loot):** Tasks waiting on parts or research.
- **Done (The Trophy Room):** Completed tasks. Reviewing this list provides a dopamine hit and validates progress (OQE).

Works cited

1. Engineering Technician 2.docx